

Scope of the Project

Traditional machine learning methods excel at identifying patterns and predicting outcomes, but they fundamentally rely on correlations. They fall short when tasked with answering "what-if" questions or evaluating the direct effect of a new intervention. In healthcare and medicine, simply knowing that two variables are correlated is not enough; researchers and clinicians need to understand the underlying mechanisms.

Causal Artificial Intelligence (AI) bridges this gap by moving beyond mere correlation [1]. It allows us to discover the true data-generating process, adjust for confounding variables, and estimate the actual impact of lifestyle changes. This project highlights what Causal AI offers that traditional predictive models cannot: the ability to reason about counterfactuals and make reliable, individualized treatment decisions from purely observational data.

Data

You will work with cross-sectional observational data from the National Health and Nutrition Examination Survey (NHANES) [2]. The dataset contains demographic records, physiological body measures, including Body Mass Index (BMI), physical activity levels, and detailed dietary nutrient intake (such as total calories, protein, and carbohydrates). The measurements represent a single snapshot of several thousand individuals.

Experiment

You will have to make yourself familiar with the fundamentals of causal discovery and inference. If you are new to these concepts, we highly recommend reviewing the Introduction to Causal Inference course [3] as a foundational learning resource before beginning technical implementation.

Your task is to build a complete causal pipeline divided into two parts:

1. **Comparative Causal Discovery:** You are free to choose any causal discovery algorithms you prefer. However, you are required to apply at least three different causal discovery methods to the dataset to learn the underlying Directed Acyclic Graph (DAG) [4]. Due to the inherent difficulty of learning causal structures from purely observational data, you may encounter conflicting structural outputs or cyclical graphs. Present the visual DAGs generated by each method and critically compare their structural differences to select the most plausible graph.
Note: for the next part, select the graph that makes sense. Some algorithms can receive fixed edges as background knowledge to force the causal discovery to include (or exclude) those edges.
2. **Causal Inference:** Using your selected causal graph, develop a robust causal inference pipeline. You must identify valid adjustment sets (confounders) and estimate the Average Treatment Effect (ATE) for multiple interventions on the final outcome, BMI. Specifically, you need to calculate the ATE for four distinct treatments:
 - Treatment A: Vigorous Physical Activity (Binary)
 - Treatment B: Total Calorie Intake (Continuous)
 - Treatment C: Protein Intake (Continuous)
 - Treatment D: Carbohydrate Intake (Continuous)

You are free to choose your preferred Python libraries (e.g., causal-learn, DoWhy) to complete this pipeline.

References

- [1] Pearl, Judea. Causality: Models, Reasoning, and Inference. Cambridge University Press, 2009.
- [2] Centers for Disease Control and Prevention (CDC). National Health and Nutrition Examination Survey (NHANES). National Center for Health Statistics. Available at: <https://www.cdc.gov/nchs/nhanes/index.htm>
- [3] Neal, Brady. Introduction to Causal Inference from a Machine Learning Perspective. Course materials available at: <https://www.bradyneal.com/causal-inference-course>
- [4] Zanga, Alessio, Elif Ozkirimli, and Fabio Stella. "A survey on causal discovery: Theory and practice." International Journal of Approximate Reasoning 151 (2022): 101-129.